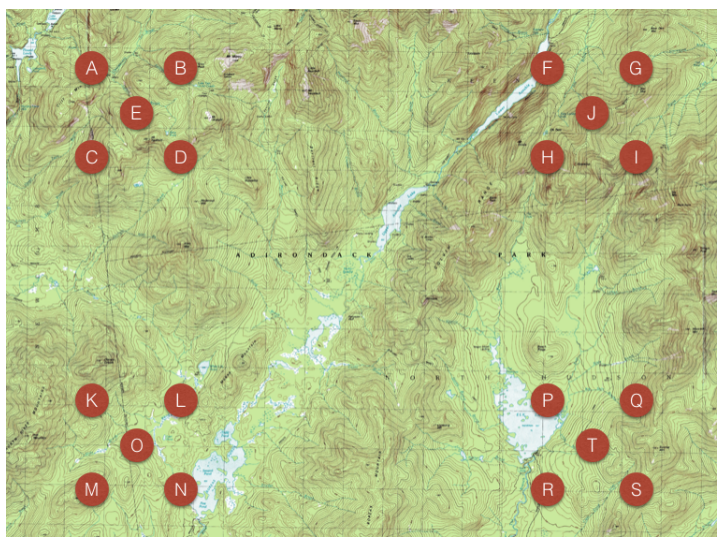
Recitation Week 13

1. For the last CS3230 mission, we will be going to war. General Lee is looking for a person who is capable of designing efficient algorithms to help plan his strategies. He heard that students who took CS3230 are capable of designing approximate algorithms. He has enlisted your help, and you cannot disobey his commands.

   To prepare for battle, General Lee has n camps spread across a flat terrain. Out of these n camps, he would like to select k camps to store medical supplies. He would like to select the camps such that the distance between any camp and the medical supplies is minimized.

   More formally, V is a set of n points spread across an Euclidean plane. The distance between any two points is given by the Euclidean distance $d(i, j)$. In addition, we define the distance between a point i and a set of points S as $d(i, S) = \min_{j \in S} d(i, j)$ (which is basically the minimum distance between any point in S and i). We would like to select a subset S of k points from V such that the quantity $R_{\max} = \max_{i \in V} d(i, S)$ is minimized. We call this quantity $R_{\max}$ the radius of V with respect to S.

   (a) The following diagram depicts the location of General Lee's camp. Select 4 points to form the set S such that $R_{\max}$ is minimized.



   Solution: We should select $S = \{E, J, O, T\}$.

(b) Consider the following greedy algorithm in generating S.

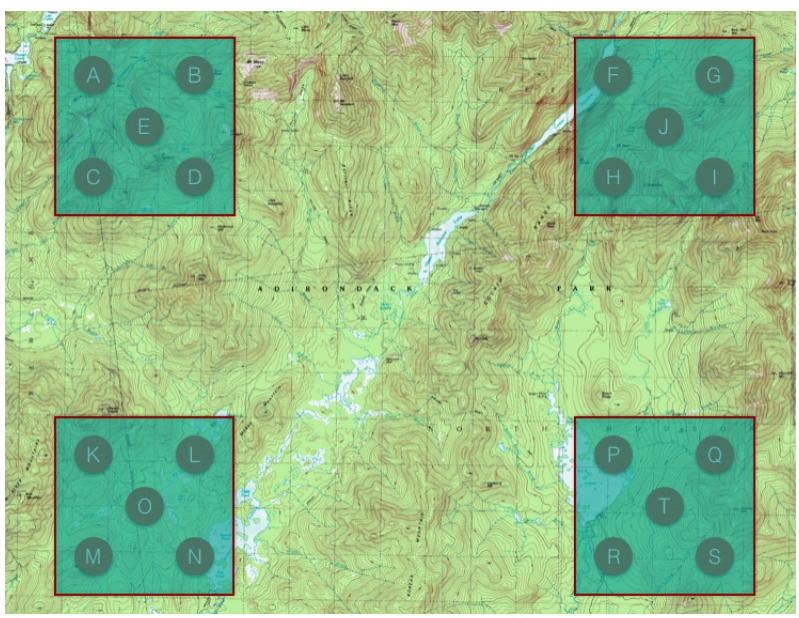
GREEDYSELECTION

Pick arbitrary $i \in V$.
$S = \{i\}$.
while $|S| < k$
    $j = \arg\max_{j \in V} d(j, S)$.
    $S \leftarrow S \cup \{j\}$.

Conduct the above algorithm to select the set S of size 4 on the map in question 1, assuming that vertex A is selected first.

Solution: The algorithm will select $S = \{A, S, G, M\}$

(c) We would like to show that the greedy algorithm suggested above is a 2-approximation algorithm for General Lee's problem. Suppose $S^* = \{j_1, j_2, \cdots, j_k\}$ is the optimal solution, and denote $R^*_{\max}$ as the radius of V with respect to S^*. Note that we can partition V into "clusters" $V_1, V_2, \cdots, V_k$ where each point $j \in V$ is placed in the cluster V_i if it is closest to $j_i \in S^*$ (we break ties arbitrarily). Refer to the following diagram for more information.



Prove that every pair of points in a cluster is at most $2R^*_{\max}$ apart.

Solution: Suppose two points $u, v \in V$ are located in the same cluster V_k that contains the point j_m. Since u belongs to V_m, we note that

$$d(u, j_m) = \min_{i=1,2,\cdots,k} d(u, j_i) = d(u, S^*) \leq R^*_{\max}$$

Similarly,

$$d(v, j_m) \leq R^*_{\max}$$

By the triangle inequality, we have

$$d(u,v) \leq d(u,j_m) + d(v,j_m) \Rightarrow d(u,v) \leq 2R^*_{\max}$$

(d) Now consider the set S selected by the greedy algorithm. Prove that all points in V are within $2R^*_{\max}$ from some point in S. Hence, the greedy algorithm returns a 2-approximate solution to General Lee's problem.

Hint: Consider what happens if GREEDYSELECTION selects k locations such that each location belongs to a different cluster of the optimal solution, and what happens if two points are selected in the same cluster during the iterations of GREEDYSELECTION.

Solution: Suppose all elements in S belongs to distinct clusters $V_1, V_2, \cdots, V_k$. Then, we note that for any $u \in V$, we have

$$d(u,S) = \min_{j \in S} d(u,j) \leq 2R^*_{\max},$$

and we are done.

Now, suppose the greedy algorithm selects two points within the same cluster. This is to say, the algorithm selected point $j' \in V_i$ after selecting point $j \in V_i$. Since j and j' belongs to the same cluster, we have $d(j,j') \leq 2R^*_{\max}$. Note that the greedy algorithm selected point j' because it satisfies

$$d(j',S') = \max_{u \in V} d(u,S')$$

where S' denotes the set of vertices selected by the greedy algorithm before j' is selected. Since $j \in S'$, we note that

$$\max_{u \in V} d(u,S') \leq d(j,j') \leq 2R^*_{max}$$

As S' is a subset of S, we note that

$$\max_{u \in V} d(u,S) \leq \max_{u \in V} d(u,S') \leq 2R^*_{\max}.$$

Hence, the greedy algorithm returns a 2-approximate solution.

Comment: It may appear as if it is not possible for the algorithm to select two points in the same cluster. This is not true. The following example shows that it can happen. Consider 4 points on a line at locations 0, 3, 5, 7. The optimal solution with two points selects locations 0 and 5 giving $R^*_{\max} = 2$ and the clusters $\{0\}$ and $\{3,5,7\}$. If location 3 is selected first by the algorithm, it will select location 7 which is in the same cluster next.